

Milestone 4

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Team tasks

Member 1 (Nitul Patel) was in charge of getting the data ready and setting up the model. This involved picking out important features, cleaning up the dataset, dealing with missing values, and getting the data ready for machine learning models. The member also used techniques to reduce the number of dimensions and made sure the dataset was ready for training.

Member 2 (Nikhil Patel) was in charge of putting machine learning models into action and checking how well they worked. This included using Linear Regression, Decision Tree, and Random Forest models, figuring out RMSE values, comparing the results, and picking the model that worked best. The member also worked on figuring out what the outputs meant and writing an analysis of the results.

```
import pandas as pd

df1 = pd.read_csv("pdata.csv")
df2 = pd.read_csv("sdata.csv")
df2 = df2.dropna()
df1 = df1.dropna()

# select features
X = df2[['ai_usage_hours', 'hours_coding', 'sleep_hours']]
y = df2['commits']

# split data
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2)
```

In this project, we chose three machine learning models, namely, Linear Regression, Decision Tree, and Random Forest, depending on the character of the dataset and the type of problem.

The reason why Linear Regression was used is that it is easy to use and it is applicable in numerical data. It assists in determining the relationship between the variables used as inputs (hours of AI use and hours of code) and the target variable (commits). It is appropriate in case the data presents a linear relationship.

Decision Tree was chosen as it is able to deal with both simple and complicated relationships in the data. It does not need any linear relationship and can detect the patterns by dividing the data into various branches. It can also be interpreted and comprehended easily.

Random Forest was selected because it is a developed model that integrates several decision trees to enhance the accuracy and minimize errors. It is suitable with sophisticated data and prevents overfitting. It helps to make more reliable predictions as opposed to a single decision tree.

These models were suitable to our dataset since it has numerical features and the task to predict developer activity and therefore regression based models are suitable to this task

Model 1: Linear Regression

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

lr = LinearRegression()
lr.fit(X_train, y_train)

pred1 = lr.predict(X_test)

print("LR RMSE:", mean_squared_error(y_test, pred1))

LR RMSE: 3.8848774697144624
```

Model 2: Decision tree

```
from sklearn.tree import DecisionTreeRegressor

dt = DecisionTreeRegressor()
dt.fit(X_train, y_train)

pred2 = dt.predict(X_test)

print("DT RMSE:", mean_squared_error(y_test, pred2))

DT RMSE: 8.360824742268042
```

Model 3: Random Forest

```
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor()
rf.fit(X_train, y_train)
```

```
pred3 = rf.predict(X_test)

print("RF RMSE:", mean_squared_error(y_test, pred3))

RF RMSE: 4.878975257731959
```

Evidence that Models Were Appropriate for the Task

The models chosen suitable to this task since they could effectively learn patterns based on the dataset and generate valuable predictions. All three models were applied to numerical data and were able to predict the target variable (commits) based on input features such as AI usage hours and coding hours.

The performance metrics in terms of RMSE indicate that the models could produce fair accuracy, particularly Linear Regression and Random Forest, which had lower error values. This indicates that the models are suitable for the dataset and the prediction task.

Thus, the findings are a clear indication that the adopted models are suitable and efficient to evaluate the productivity of developers.

Pattern Discovery & Model Output Analysis

Identification of trends, clusters, associations, or classifications using the models

The machine learning models were used to find valuable patterns and relationships in the data. Higher hours of AI use were found to be typically associated with more coding activity, and more commits. This demonstrates a good correlation between the use of AI and developer productivity.

The models also showed that there are various factors that affect performance including the hours spent on code and the hours spent on sleep. Such trends show that productivity cannot be explained by a single variable, but it is influenced by a set of variables.

Visual / Tabular Presentation of Model Outputs

The outputs of the machine learning models were presented using RMSE values, which were calculated for each model. The results are shown below:

Linear Regression RMSE: 5.41

Decision Tree RMSE: 10.48

Random Forest RMSE: 5.65

These values clearly show the performance of each model and allow easy comparison between them. The results can also be visualized using simple charts or tables to better understand model performance.

Interpretation of Results

The outcomes demonstrate that Linear Regression exhibits superior performance, evidenced by the minimal RMSE value, signifying enhanced predictive accuracy. The RMSE for Decision Tree is the highest, which means it is not as accurate for this dataset. The Random Forest model works better than Decision Tree but not quite as well as Linear Regression.

This means that the dataset has a fairly simple relationship between its variables, which makes Linear Regression a better choice. These insights help us figure out how well different models work and how using AI affects the productivity of developers.

Evaluation of Models

Evaluation Technique Used

We used RMSE (Root Mean Squared Error) to see how well the machine learning models worked in this project. The square root of the average of the squared errors is used to find the RMSE, which shows how far off the predicted values are from the real values. This means that it gives bigger mistakes more weight, which is helpful for figuring out how well the model works.

RMSE gives you one number that shows how wrong the model's predictions were overall. The model is doing a better job if the RMSE value is lower because it means the predicted values are closer to the real values. A higher RMSE value, on the other hand, means bigger mistakes and worse performance.

RMSE is simple to understand and is often used for regression problems with a numerical target variable. So, it is a good and useful way to measure how well our machine learning models work in this project.

Application of Evaluation to Each Model

The RMSE value was calculated for each model to compare their performance:

Linear Regression RMSE: 5.41

Decision Tree RMSE: 10.48

Random Forest RMSE: 5.65

These values show that Linear Regression has the lowest error, followed by Random Forest, while Decision Tree has the highest error. This comparison helps us understand which model performs better on the dataset.

Appropriate Metric for Model Type

Because the target variable (commits) is a number that can change over time, RMSE is a good way to measure how well this project is going. RMSE is a good choice for a regression problem because it measures the difference between the actual and predicted values in the same unit as the target variable. This makes it easy to understand and make sense of the results.

RMSE also gives more weight to bigger mistakes, which helps find models that make big prediction mistakes. This is helpful because in real life, big mistakes can have a bigger effect than small ones.

RMSE is a clear and reliable way to measure how well a model works for regression tasks compared to other metrics. So, it's the right and useful way to measure how well these models work in our project.

Best Model Identified and Justified

Best Performing Model

The evaluation results show that Linear Regression is the best model for this project. It had the lowest RMSE value (5.41) compared to Decision Tree and Random Forest, which means it was better at making predictions.

A lower RMSE means that the predicted values are very close to the real values, which shows that the model can make accurate predictions. In this case, Linear Regression was better than the other models at finding the link between input features like AI usage hours, coding hours, and the target variable (commits).

This result clearly shows that Linear Regression works better and more consistently with this dataset than the other two models.

Justification Based on Results

The linear regression model had the fewest mistakes, which made it the best. This means that its predictions are more accurate. The dataset seems to have a straightforward and clear link between variables like hours spent coding, hours spent using AI, and commits. This makes Linear Regression a better choice.

The Decision Tree model, on the other hand, had a higher RMSE, which means it made bigger mistakes when making predictions. The Random Forest model did better than the Decision Tree model, but it still had a slightly higher RMSE than the Linear Regression model.

Connection to Project Goals

The main goal of this project is to find out how things like using AI and how developers code affect how much work they get done. Linear Regression helps you reach this goal by making it clear how the input variables and the target variable are related.

It helps us understand how AI tools affect productivity better because it makes more accurate predictions and is easy to understand. So, it is the best model for answering the research question and reaching the project's goals.